



S.P.M College, Udantpuri

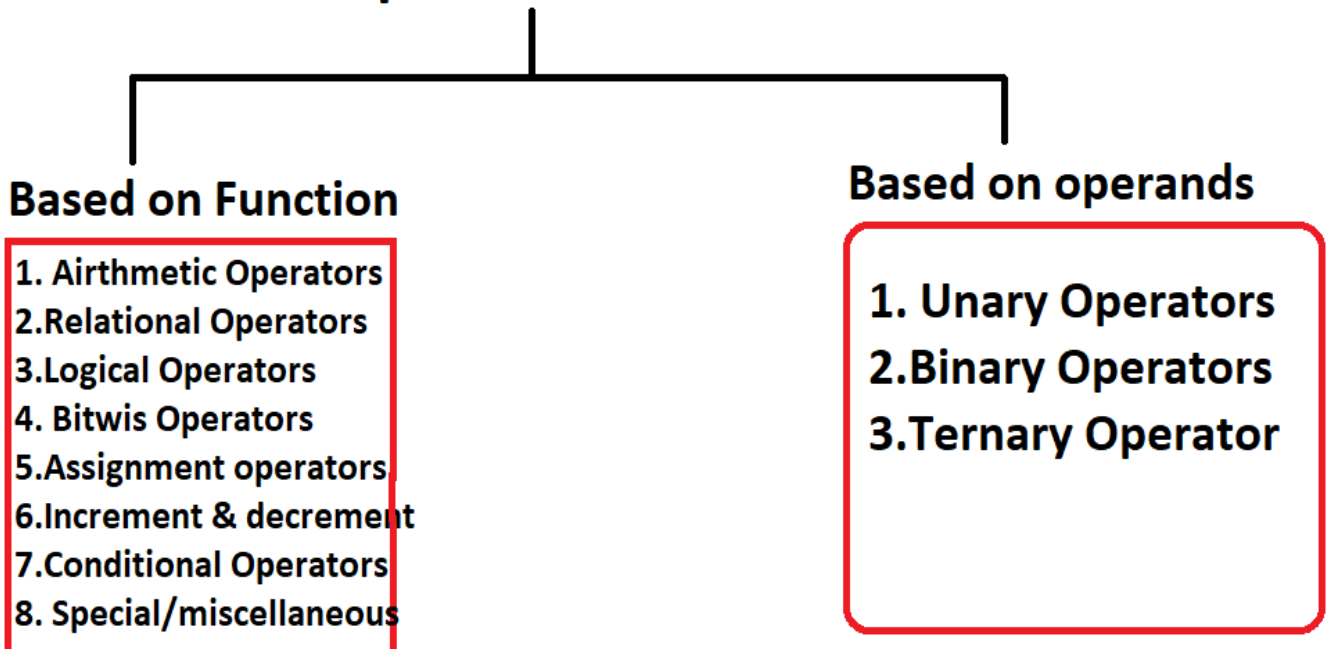
Bachelor Of Computer Application (BCA)

Part -2 (Paper-3)

– Hira Kumar

Operators in C++

Operators in c++



➤ Based on Operands

There are three categories of operators in C++, these are-

- 1. Unary Operators**
- 2. Binary Operators**
- 3. Ternary Operators**

Operators in C

	Operator	Type
Unary operator	++, --	Unary operator
Binary operator	+, -, *, /, %	Arithmetic operator
	<, <=, >, >=, ==, !=	Relational operator
	&&, , !	Logical operator
	&, , <<, >>, ~, ^	Bitwise operator
	=, +=, -=, *=, /=, %=	Assignment operator
Ternary operator	?:	Ternary or conditional operator

1. Unary Operators

The operators that operate a single operand to form an expression are known as unary operator. The operators like ++ (increment) operator, -- (decrement) operator etc. are the part of unary operators.

Ex → ++a, b--,....etc

2. Binary Operators

The operators that operate two or more operands are known as binary operators. The operators like +, -, *, /, % etc are binary operators.

Ex → a+b, a-b, a*b, a/b etc

3. Ternary Operators

The operator that operates minimum or maximum three operands is known as ternary operator. there is only one ternary operator available in C++. The operator ?: is the only available ternary operator i.e used as a substitute of if-else statement.

Ex → a>b?a:b

<<	insertion operator
>>	extraction operator
::	scope resolution operator
::*	pointer to member declarator
*	pointer to member operator
.*	pointer to member operator
Delete	memory release operator
endl	line feed operator
New	memory allocation operator
setw	field width operator

Home work - Formula-Based Programs

1. Write a C++ program to find the Simple Interest. (*Formula: $SI = (P \times R \times T) / 100$*)
2. Write a C++ program to find the Compound Interest. (*Formula: $CI = P(1 + R/100)^T - P$*)
3. Write a C++ program to calculate the Area of a Rectangle. (*Formula: $A = L \times B$*)
4. Write a C++ program to calculate the Perimeter of a Rectangle. (*Formula: $P = 2(L + B)$*)
5. Write a C++ program to calculate the Area of a Square. (*Formula: $A = a^2$*)
6. Write a C++ program to calculate the Perimeter of a Square. (*Formula: $P = 4a$*)
7. Write a C++ program to calculate the Area of a Triangle. (*Formula: $A = \frac{1}{2} \times b \times h$*)
8. Write a C++ program to calculate the Area of a Circle. (*Formula: $A = \pi r^2$*)
9. Write a C++ program to calculate the Circumference of a Circle. (*Formula: $C = 2\pi r$*)

10. Write a C++ program to calculate the Area of a Parallelogram.
(Formula: $A = b \times h$)
11. Write a C++ program to calculate the Area of a Trapezium.
(Formula: $A = \frac{1}{2}(a + b) \times h$)
12. Write a C++ program to calculate the Area of a Rhombus.
(Formula: $A = \frac{1}{2} \times d_1 \times d_2$)
13. Write a C++ program to calculate the Area of an Equilateral Triangle. (Formula: $A = (\sqrt{3}/4)a^2$)
14. Write a C++ program to calculate the Volume of a Cube.
(Formula: $V = a^3$)
15. Write a C++ program to calculate the Total Surface Area of a Cube. (Formula: $TSA = 6a^2$)
16. Write a C++ program to calculate the Lateral Surface Area of a Cube. (Formula: $LSA = 4a^2$)
17. Write a C++ program to calculate the Volume of a Cuboid.
(Formula: $V = L \times B \times H$)
18. Write a C++ program to calculate the Total Surface Area of a Cuboid. (Formula: $TSA = 2(LB + BH + HL)$)
19. Write a C++ program to calculate the Lateral Surface Area of a Cuboid. (Formula: $LSA = 2H(L + B)$)
20. Write a C++ program to calculate the Volume of a Cylinder.
(Formula: $V = \pi r^2 h$)
21. Write a C++ program to calculate the Curved Surface Area of a Cylinder. (Formula: $CSA = 2\pi r h$)
22. Write a C++ program to calculate the Total Surface Area of a Cylinder. (Formula: $TSA = 2\pi r(h + r)$)
23. Write a C++ program to calculate the Volume of a Cone.
(Formula: $V = (1/3)\pi r^2 h$)
24. Write a C++ program to calculate the Curved Surface Area of a Cone. (Formula: $CSA = \pi r l$)
25. Write a C++ program to calculate the Total Surface Area of a Cone. (Formula: $TSA = \pi r(l + r)$)
26. Write a C++ program to calculate the Volume of a Sphere.
(Formula: $V = (4/3)\pi r^3$)
27. Write a C++ program to calculate the Surface Area of a Sphere.
(Formula: $SA = 4\pi r^2$)
28. Write a C++ program to calculate the Volume of a Hemisphere.
(Formula: $V = (2/3)\pi r^3$)

29. Write a C++ program to calculate the Total Surface Area of a Hemisphere. (*Formula: $TSA = 3\pi r^2$*)
30. Write a C++ program to calculate the Curved Surface Area of a Hemisphere. (*Formula: $CSA = 2\pi r^2$*)
31. Write a C++ program to calculate the Diagonal of a Rectangle. (*Formula: $d = \sqrt{L^2 + B^2}$*)
32. Write a C++ program to calculate the Diagonal of a Square. (*Formula: $d = a\sqrt{2}$*)
33. Write a C++ program to calculate the Hypotenuse of a Right Triangle. (*Formula: $c = \sqrt{a^2 + b^2}$*)
34. Write a C++ program to calculate the Distance Between Two Points. (*Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$*)
35. Write a C++ program to calculate the Percentage. (*Formula: $Percentage = (Obtained\ Marks / Total\ Marks) \times 100$*)
36. Write a C++ program to calculate the Profit or Loss. (*Formula: $Profit = SP - CP$, $Loss = CP - SP$*)
37. Write a C++ program to calculate the Discount Amount and Final Price. (*Formula: $Discount = (MP \times Discount\%) / 100$, $Final\ Price = MP - Discount$*)
38. Write a C++ program to calculate the Speed. (*Formula: $Speed = Distance / Time$*)
39. Write a C++ program to calculate the Distance. (*Formula: $Distance = Speed \times Time$*)
40. Write a C++ program to calculate the Time. (*Formula: $Time = Distance / Speed$*)

C++ Formula Based Programs (Mensuration & Other Formulas)

1. Simple Interest ($SI = \frac{P \times R \times T}{100}$)	21. Curved Surface Area of Cylinder ($CSA = 2\pi r h$)
2. Compound Interest ($CI = P(1 + \frac{R}{100})^T - P$)	22. Total Surface Area of Cylinder ($TSA = 2\pi r(h + r)$)
3. Area of Rectangle ($A = L \times B$)	23. Volume of Cone ($V = \frac{1}{3}\pi r^2 h$)
4. Perimeter of Rectangle ($P = 2(L + B)$)	24. Curved Surface Area of Cone ($CSA = \pi r l$)
5. Area of Square ($A = a^2$)	25. Total Surface Area of Cone ($TSA = \pi r(l + r)$)
6. Perimeter of Square ($P = 4a$)	26. Volume of Sphere ($V = \frac{4}{3}\pi r^3$)
7. Area of Triangle ($A = \frac{1}{2} \times b \times h$)	27. Surface Area of Sphere ($SA = 4\pi r^2$)
8. Area of Circle ($A = \pi r^2$)	28. Volume of Hemisphere ($V = \frac{2}{3}\pi r^3$)
9. Circumference of Circle ($C = 2\pi r$)	29. Total Surface Area of Hemisphere ($TSA = 3\pi r^2$)
10. Area of Parallelogram ($A = b \times h$)	30. Curved Surface Area of Hemisphere ($CSA = 2\pi r^2$)
11. Area of Trapezium ($A = \frac{1}{2}(a + b) \times h$)	31. Diagonal of Rectangle ($d = \sqrt{L^2 + B^2}$)
12. Area of Rhombus ($A = \frac{1}{2} \times d_1 \times d_2$)	32. Diagonal of Square ($d = a\sqrt{2}$)
13. Area of Equilateral Triangle ($A = \frac{\sqrt{3}}{4} a^2$)	33. Hypotenuse of Right Triangle ($C = \sqrt{a^2 + b^2}$)
14. Volume of Cube ($V = a^3$)	34. Distance Between Two Points ($d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$)
15. Total Surface Area of Cube ($TSA = 6a^2$)	35. Percentage ($\text{Percentage} = \frac{\text{Obtained Marks}}{\text{Total Marks}} \times 100$)
16. Lateral Surface Area of Cube ($LSA = 4a^2$)	36. Profit or Loss ($\text{Profit} = SP - CP, \text{Loss} = CP - SP$)
17. Volume of Cuboid ($V = L \times B \times H$)	37. Discount Amount & Final Price ($\text{Discount} = \frac{MP \times \text{Discount \%}}{100}, \text{Final Price} = MP - \text{Discount}$)
18. Total Surface Area of Cuboid ($TSA = 2(LB + BH + HL)$)	38. Speed ($\text{Speed} = \frac{\text{Distance}}{\text{Time}}$)
19. Lateral Surface Area of Cuboid ($LSA = 2H(L + B)$)	39. Distance ($\text{Distance} = \text{Speed} \times \text{Time}$)
20. Volume of Cylinder ($V = \pi r^2 h$)	40. Time ($\text{Time} = \frac{\text{Distance}}{\text{Speed}}$)

MENSURATION

PART A : 2D MENSURATION

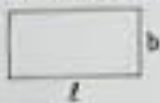
1. SQUARE



Side = a

$$\begin{aligned}\text{Perimeter} &= 4a \\ \text{Area} &= a^2 \\ \text{Diagonal} &= a\sqrt{2}\end{aligned}$$

2. RECTANGLE



Length = ℓ , Breadth = b

$$\begin{aligned}\text{Perimeter} &= 2(\ell + b) \\ \text{Area} &= \ell \times b \\ \text{Diagonal} &= \sqrt{\ell^2 + b^2}\end{aligned}$$

3. TRIANGLE



$$\text{Area} = \frac{1}{2} \times b \times h$$

Heron's Formula :

$$s = \frac{a+b+c}{2}$$

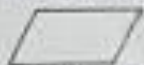
$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

4. EQUILATERAL TRIANGLE Side = a



$$\begin{aligned}\text{Area} &= \frac{\sqrt{3}}{4} a^2 \\ \text{Height} &= \frac{\sqrt{3}}{2} a\end{aligned}$$

5. PARALLELOGRAM



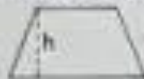
$$\text{Area} = \text{Base} \times \text{Height}$$

6. RHOMBUS



$$\text{Area} = \frac{1}{2} \times d_1 \times d_2$$

7. TRAPEZIUM



$$\text{Area} = \frac{1}{2} \times (\text{Sum of parallel sides}) \times \text{Height}$$

8. CIRCLE



Radius = r

$$\begin{aligned}\text{Circumference} &= 2\pi r \\ \text{Area} &= \pi r^2\end{aligned}$$

$$\text{Area of semicircle} = \frac{1}{2} \pi r^2$$

$$\text{Perimeter of semicircle} = \pi r + 2r$$

$$\text{Area of quadrant} = \frac{1}{4} \pi r^2$$

9. SECTOR OF CIRCLE



$$\begin{aligned}\text{Area} &= \frac{\theta}{360} \times \pi r^2 \\ \text{Arc length} &= \frac{\theta}{360} \times 2\pi r\end{aligned}$$

PART B : 3D MENSURATION

10. CUBE



Side = a

$$\begin{aligned}\text{Volume} &= a^3 \\ \text{LSA} &= 4a^2 \\ \text{TSA} &= 6a^2 \\ \text{Diagonal} &= a\sqrt{3}\end{aligned}$$

11. CUBOID



Length = ℓ , Breadth = b , Height = h

$$\begin{aligned}\text{Volume} &= lbh \\ \text{LSA} &= 2h(\ell + b) \\ \text{TSA} &= 2(lb + bh + lh) \\ \text{Diagonal} &= \sqrt{\ell^2 + b^2 + h^2}\end{aligned}$$

12. CYLINDER



Radius = r , Height = h

$$\begin{aligned}\text{Volume} &= \pi r^2 h \\ \text{CSA} &= 2\pi rh \\ \text{TSA} &= 2\pi r(r + h)\end{aligned}$$

13. CONE



Radius = r
Height = h
Slant height ℓ
 $= \sqrt{r^2 + h^2}$

$$\begin{aligned}\text{Volume} &= \frac{1}{3} \pi r^2 h \\ \text{CSA} &= \pi r \ell \\ \text{TSA} &= \pi r(\ell + r)\end{aligned}$$

14. SPHERE



Radius = r

$$\begin{aligned}\text{Volume} &= \frac{4}{3} \pi r^3 \\ \text{Surface Area} &= 4\pi r^2\end{aligned}$$

15. HEMISPHERE



$$\begin{aligned}\text{Volume} &= \frac{2}{3} \pi r^3 \\ \text{CSA} &= 2\pi r^2 \\ \text{TSA} &= 3\pi r^2\end{aligned}$$

16. FRUSTUM OF CONE



R = bigger radius, r = smaller radius

$$\begin{aligned}\text{Volume} &= \frac{1}{3} \pi h(R^2 + r^2 + Rr) \\ \text{CSA} &= \pi \ell(R + r) \\ \text{TSA} &= \pi \ell(R + r) + \pi(R^2 + r^2)\end{aligned}$$

17. IMPORTANT RATIO TRICKS

- If radius doubles $\rightarrow$ Area $\times 4$, Volume $\times 8$
- If dimensions scaled by $k \rightarrow$ Area $\times k^2$, Volume $\times k^3$

18. CONVERSION OF SOLIDS

Volume before = Volume after

(Q) Cone melted into sphere

$\Rightarrow$ Equate volumes

19. IMPORTANT SHORTCUTS

- ✓ Always write dimensions first
- ✓ Use $\pi = \frac{22}{7}$ or 3.14 carefully
- ✓ Convert $\text{cm}^3 \rightarrow \text{m}^3$ if required
- ✓ Volume conserved in melting

20. COMMON MISTAKES

- ✗ Mixing CSA & TSA
- ✗ Forgetting slant height
- ✗ Unit conversion errors
- ✗ Wrong π value
- ✗ Mixing CSA & TSA
- ✗ Forgetting slant height
- ✗ Unit conversion errors
- ✗ Wrong π value

=====HIRA KUMAR=====